$$\therefore \qquad dE = -kT\,dx$$

For $E = E_v$, $x = 0$

$$\therefore \quad p = \frac{4\pi}{h^3}\left(2m_p^*\right)^{3/2}\exp\left(\frac{E_v - E_F}{kT}\right)\int_{\infty}^{0} x^{1/2}(kT)^{1/2}e^{-x}(-kT)\,dx$$

$$= \frac{4\pi}{h^3}\left(2m_p^*\right)^{3/2}\exp\left(\frac{E_v - E_F}{kT}\right)(kT)^{3/2}\int_{0}^{\infty} x^{1/2}e^{-x}\,dx$$

$$= \frac{4\pi}{h^3}\left(2m_p^*kT\right)^{3/2}\exp\left(\frac{E_v - E_F}{kT}\right)\left(\frac{\pi}{4}\right)^{1/2}$$

$$= 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}\exp\left[-\left(\frac{E_F - E_v}{kT}\right)\right] \tag{7.25}$$

or
$$p = N_v \exp\left[-\left(\frac{E_F - E_v}{kT}\right)\right] \tag{7.26}$$

where

$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} \tag{7.27}$$

represents the effective density of holes at the valence band edge. For silicon,

$$N_v = 2.8 \times 10^{25}\left(\frac{T}{300}\right)^{3/2} m^{-3}$$

The electron and hole concentrations given by Eqs. (7.20) and (7.26) respectively are valid for both intrinsic and extrinsic materials. For intrinsic materials, these equations can also be written as

$$n_i = N_c \exp\left[-\left(\frac{E_c - E_i}{kT}\right)\right], \quad p_i = N_v \exp\left[-\left(\frac{E_i - E_v}{kT}\right)\right] \tag{7.28}$$

where the Fermi level E_F has been replaced by the intrinsic level E_i.

7.5.3 Fermi Level

For an intrinsic semiconductor,

$$n = p = n_i$$

Therefore, from Eqs. (7.20) and (7.26), we get

$$N_c \exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right] = N_v \exp\left[-\left(\frac{E_F - E_v}{kT}\right)\right]$$

or
$$\exp\left(\frac{2E_F - E_c - E_v}{kT}\right) = \frac{N_v}{N_c}$$

or
$$\frac{2E_F - E_c - E_v}{kT} = \ln\left(\frac{N_v}{N_c}\right)$$

or
$$E_F = E_i = \frac{E_c + E_v}{2} + \frac{kT}{2} \ln\left(\frac{N_v}{N_c}\right) \tag{7.29}$$

Using Eqs. (7.21) and (7.27) in Eq. (7.29), we obtain

$$E_F = \frac{E_c + E_v}{2} + \frac{3}{4} kT \ln\left(\frac{m_p^*}{m_n^*}\right) \tag{7.30}$$

At 0 K,

$$E_F = \frac{E_c + E_v}{2} \tag{7.31}$$

i.e., the Fermi level lies in the middle of the conduction band and valence band. This is also true at all other temperatures provided $m_p^* = m_n^*$. However, in general, $m_p^* > m_n^*$ and the Fermi level is raised slightly as T exceeds 0 K. For Si at 300 K, the increase in Fermi energy is about 0.01 eV only which may be neglected for all practical purposes.

7.5.4 Law of Mass Action and Intrinsic Carrier Concentration

Since for an intrinsic semiconductor,

$$n = p = n_i,$$

the Eqs. (7.20) and (7.26) yield

$$np = n_i^2 = N_c N_v \exp\left(-\frac{E_c - E_v}{kT}\right)$$

$$= N_c N_v \exp\left(-\frac{E_g}{kT}\right) \tag{7.32}$$

Using Eqs. (7.21) and (7.27) , the above expression becomes

$$np = n_i^2 = 4\left(\frac{2\pi kT}{h^2}\right)^3 \left(m_n^* m_p^*\right)^{3/2} e^{-E_g/kT} \tag{7.33}$$

This shows that, for a given semiconductor, the product of electron and hole concentrations is a constant at a given temperature and is equal to the square of the intrinsic carrier concentration. This is called the *law of mass action* and holds for both intrinsic and extrinsic semiconductors. If impurity atoms are added to a semiconductor to increase n, there will be a corresponding decrease in p such that the product np remains constant. Thus we always have

$$np = n_i^2 \tag{7.34}$$

The intrinsic carrier concentration can be directly obtained from Eq. (7.32) or (7.33) as

$$n_i = (N_c N_v)^{1/2} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.35}$$

$$= 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} \left(m_n^* m_p^*\right)^{3/4} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.36}$$

For pure Ge at 300 K, the intrinsic electron concentration is about 2.4×10^{19} m^{-3} when the concentration of germanium atoms is 4.4×10^{28} m^{-3}. This shows that, at ordinary temperatures, only about five covalent bonds per 10^{10} atoms of germanium are broken and contribute to intrinsic conduction. On the other hand, in metals such as copper, about 10^{28} electrons per cubic metre are available for conduction.

7.6 CARRIER CONCENTRATION, FERMI LEVEL AND CONDUCTIVITY FOR EXTRINSIC SEMICONDUCTOR

As described earlier, the donor or acceptor levels are present in an extrinsic semiconductor depending on the type of impurity present. The concentration of donors or acceptors in a semiconductor affects the Fermi energy, the carrier concentration and the conductivity of the semiconductor. We consider the following cases :

(a) N-type Semiconductor

The energy level diagram for an n-type semiconductor is shown in Fig. 7.7. At 0 K, all the donors are in the unionized state, i.e., all the donor levels are occupied with electrons. As the temperature increases slightly, some of the donors get ionized and contribute electrons to the conduction band. Also, some of the valence band electrons may jump to the conduction

band leaving behind holes in the valence band. The number of holes produced in this process is, however, quite small. Therefore, the Fermi level must lie somewhere near the middle of the donor level and the bottom of the conduction band. We determine the equilibrium carrier concentration at a temperature T. Let there be N_d donors per unit volume occupying the donor levels with energy E_d. Assuming that E_F lies more than few kT below the conduction band, the electron concentration in the conduction band, as given by Eq. (7.20), is

$$n = N_c \exp\left(-\frac{E_c - E_F}{kT}\right) \tag{7.20}$$

The electron concentration must be equal to the sum of the concentration of ionized donors, N_d^+, in the donor levels and the concentration of thermally generated holes in the valence band, i.e.,

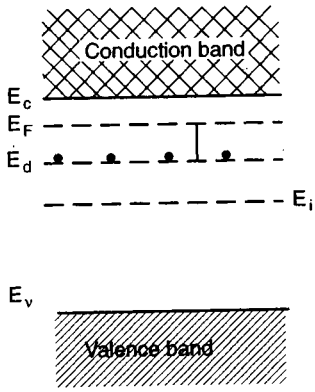


Fig. 7.7. *Band model of an n-type semiconductor.*

$$n = N_d^+ + p \tag{7.37}$$

If a sufficient number of donors are present to produce electrons in the conduction band, the concentration of thermally generated holes gets suppressed as a consequence of the law of mass action [Eq. (7.34)]. Thus p may be neglected in Eq. (7.37) which, therefore, becomes

$$n = N_d^+ \tag{7.38}$$

Also, from Eq.(7.34), the concentration of minority carriers (holes) is given by

$$p = \frac{n_i^2}{N_d^+} \tag{7.39}$$

and the concentration of ionized donors is calculated as

$$N_d^+ = N_d\,[1 - f(E_d)]$$

$$= N_d \left[1 - \frac{1}{1 + \exp\left\{-\left(\frac{E_F - E_d}{kT}\right)\right\}}\right]$$

$$= N_d \left[\frac{\exp\left\{-\left(\frac{E_F - E_d}{kT}\right)\right\}}{1+\exp\left\{-\left(\frac{E_F - E_d}{kT}\right)\right\}} \right]$$

$$= N_d \exp\left[-\left(\frac{E_F - E_d}{kT}\right)\right] \tag{7.40}$$

where we have neglected the exponent term in the denominator assuming that E_F lies more than a few kT above E_d. Using Eqs. (7.20) and (7.40) in Eq. (7.38), we obtain

$$N_c \exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right] = N_d \exp\left[-\left(\frac{E_F - E_d}{kT}\right)\right]$$

or $$\ln N_c - \left(\frac{E_c - E_F}{kT}\right) = \ln N_d - \left(\frac{E_F - E_d}{kT}\right)$$

or $$-\frac{E_c - E_F}{kT} + \frac{E_F - E_d}{kT} = \ln N_d - \ln N_c$$

or $$E_F = \frac{E_d + E_c}{2} + \frac{kT}{2} \ln\left(\frac{N_d}{N_c}\right) \tag{7.41}$$

This gives the position of the Fermi level at moderate temperatures. Since N_c varies as $T^{3/2}$ as given by Eq. (7.21), this equation is not valid for T = 0 K, Also, it is not valid for $T \to \infty$ because, at very high temperatures, the assumption of suppressing the holes does not hold good. The only valid information obtainable from this equation is that the Fermi level lies somewhere near the middle of the donor level and the conduction band edge at moderate temperatures. This is particularly true for those values of T and N_d for which the second term on the right hand side is negligible. As T increases, the Fermi level moves downwards and crosses the donor level. For sufficiently large temperatures, it drops to $E_g/2$, i.e., coincides with the intrinsic level E_i. This is, however, not apparent from Eq. (7.41). In such a case, the extrinsic semiconductor behaves like an intrinsic one. The variations of $(E_F - E_i)$ with temperature for both *n* and *p* type silicon are shown in Fig. 7.8 for different impurity concentrations.

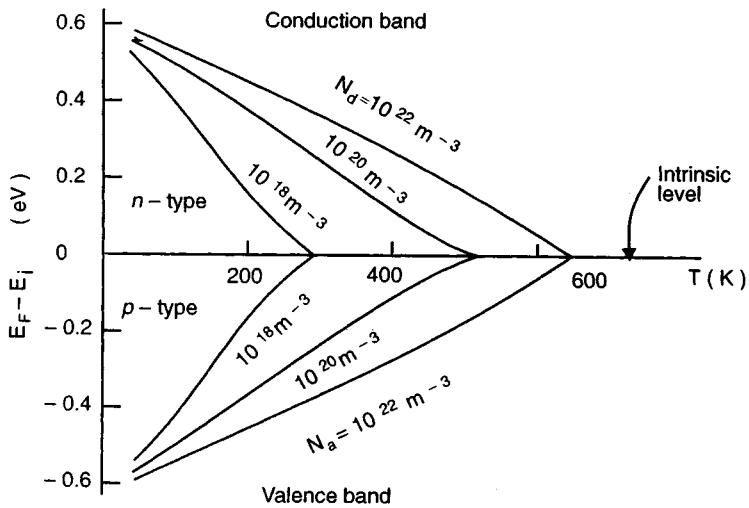


Fig. 7.8. *Variations of Fermi level with temperature and impurity concentration for Si.*

A direct relationship between the equilibrium carrier concentration and the position of the Fermi level relative to the intrinsic level, as obtained from Eqs. (7.20) and (7.28), is

$$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right) \tag{7.42}$$

It is obvious from this equation that the equilibrium electron concentration approaches the intrinsic carrier concentration as E_F approaches E_i. Also, the electron concentration increases exponentially as the Fermi level moves away from E_i towards the conduction band. A similar relation for the hole concentration will be obtained later.

The free electron concentration in the conduction band is obtained by substituting the value of E_F from Eq. (7.41) into (7.20), i.e.,

$$n = N_c \exp\left[\frac{E_d - E_c}{2kT} + \frac{1}{2}\ln\left(\frac{N_d}{N_c}\right)\right]$$

$$= N_c \left(\frac{N_d}{N_c}\right)^{1/2} \exp\left(\frac{E_d - E_c}{2kT}\right)$$

$$= (N_d N_c)^{1/2} \exp\left(\frac{E_d - E_c}{2kT}\right) \tag{7.43}$$

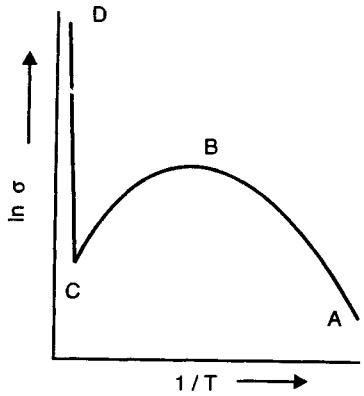


Fig. 7.9. *Conductivity versus temperature for a typical n-type germanium sample.*

$$= (N_d N_c)^{1/2} \exp\left(-\frac{\Delta E}{2kT}\right) \qquad (7.44)$$

where $\Delta E = E_c - E_d$ represents the ionization energy of the donors. This shows that the carrier concentration at moderate temperatures varies as $\sqrt{N_d}$.

The electrical conductivity of an *n*-type semiconductor can be calculated from

$$\sigma = en\mu_n \cong e\, N_d^+\, \mu_n \qquad (7.45)$$

where it has been assumed that the conduction is mainly due to electrons. Knowing the electron concentration, n, and the electron mobility, μ_n, at any given temperature, the electrical conductivity can be calculated by using Eq. (7.45). The temperature dependence of electrical conductivity for an *n*-type germanium is shown in Fig. 7.9. The following conclusions can be drawn from this curve :

(i) Starting from the very low temperature (about 10 K) corresponding to the point A, the conductivity increases with rise in temperature. This is due to the increase in the number of conduction electrons as a result of ionization of the donors. The conductivity attains a maximum value (point B) when all the donors are ionized. The temperature corresponding to this is about 50 K for a moderately doped *n*-type germanium.

(ii) The conductivity decreases with further increase in temperature up to about room temperature (point C) and is attributed to the decrease in the value of mobility with rise in temperature. There is also an increase in the intrinsic conductivity but to a lesser extent.

(iii) The sharp rise in conductivity from C to D is due to the large increase in intrinsic conductivity which offsets the decrease in mobility.

(b) P-type Semiconductor

The case of a *p*-type semiconductor can be treated in the same way as that of an *n*-type semiconductor. The energy level diagram for a *p*-type semiconductor is shown in Fig. 7.10. The acceptor impurity atoms occupy the acceptor levels, E_a, which lie above the valence band. For T > 0 K, a

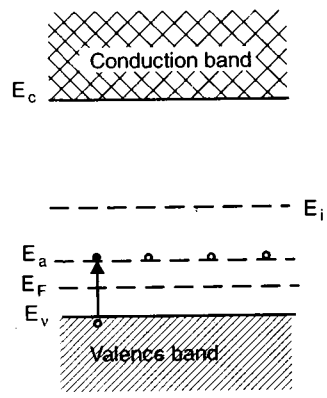


Fig. 7.10. *Band model for a p-type semiconductor.*

part of these acceptors is ionized by acquiring electrons from the valence band, thus creating holes in the valence band which cause *p*-type conduction. Apart from these holes, some thermally generated holes are also present in the valence band. If n, p, N_a and N_a^- represent electron concentration in the conduction band, hole concentration in the valence band, total acceptor concentration, and concentration of ionized acceptors respectively, then

$$p = n + N_a^-$$

Neglecting n in comparison with N_a^- for a *p*-type semiconductor, we get

$$p \cong N_a^- \tag{7.46}$$

The concentration of ionized acceptors is given by

$$N_a^- = N_a f(E_a)$$

$$= \frac{N_a}{1+\exp\left(\frac{E_a - E_F}{kT}\right)} \tag{7.47}$$

Assuming $(E_a - E_F)$ to be large as compared to kT, Eq. (7.47) can be written as

$$N_a^- \cong N_a \exp\left(\frac{E_F - E_a}{kT}\right) \tag{7.48}$$

The hole concentration in the valence band is given by Eq. (7.26) as

$$p = N_v \exp\left(-\frac{E_F - E_v}{kT}\right) \tag{7.26}$$

Using Eqs. (7.48) and (7.26) in Eq. (7.46), we obtain

$$N_v \exp\left(-\frac{E_F - E_v}{kT}\right) = N_a \exp\left(\frac{E_F - E_a}{kT}\right)$$

which, on simplification, gives

$$E_F = \frac{E_a + E_v}{2} - \frac{kT}{2} \ln\left(\frac{N_a}{N_v}\right) \tag{7.49}$$

As described earlier, the Fermi level at moderate temperatures lies near the middle of the acceptor level and the top of the valence band. It moves upwards with increase in temperature and finally coincides with the intrinsic level as shown in Fig. 7.8. The expression (7.49) is not valid at very high temperatures where n cannot be neglected.

Using Eqs. (7.26) and (7.28), we obtain an expression identical to Eq. (7.42), i.e.,

$$p = n_i \exp\left(\frac{E_i - E_F}{kT}\right) \tag{7.50}$$

It indicates that the equilibrium hole concentration approaches the intrinsic carrier concentration as E_F approaches E_i. Also, p increases exponentially as E_F moves away from E_i towards the valence band. An expression for the hole concentration in the valence band can be obtained by using Eq. (7.49) into (7.26) as in the previous case.

The electrical conductivity of a p-type semiconductor is given by

$$\sigma_p = ep\mu_p \cong eN_a^- \mu_p \tag{7.51}$$

where the conduction due to minority carriers has been ignored. The variation of conductivity with temperature is similar to that for the n-type semiconductor.

It also follows from Eqs. (7.45) and (7.51) that the increase in concentration of either type of impurity atoms increases the conductivity of a semiconductor. For impurity concentration ranging from 10^{20} to 10^{22} m^{-3}, the resistivity of Si and Ge varies from 10^{-3} to 10^{-1} ohm-m. If doping is heavy (10^{23} to 10^{24} atoms/m^3), the conductivity of semiconductors becomes comparable to metals. Such semiconductors are called *degenerate semiconductors* and find applications in high power and high frequency devices.

(c) Mixed Semiconductor

In a semiconductor containing both n and p-type impurities, the law of electrical neutrality is written as

$$N_d^+ + p = N_a^- + n \tag{7.52}$$

Taking $N_d^+ = N_a^-$ and using Eq. (7.34), we obtain

$$p = n = n_i$$

This shows that, for equal concentrations of donors and acceptors, the semiconductor behaves as pure or intrinsic semiconductor. All the ionized

acceptors combine with free electrons of the donors and all the ionized donors combine with free holes of the acceptors to produce no net free carriers. For $N_d^+ \neq N_a^-$ or simply $N_d \neq N_a$ (donors and acceptors are assumed to be ionized), the semiconductor behaves as n-type or p-type depending on the relative magnitudes of N_d and N_a, and the cases described above become applicable.

SOLVED EXAMPLES

Example 7.1. The electron and hole mobilities in a Si sample are 0.135 and 0.048 m^2/V-s respectively. Determine the conductivity of intrinsic Si at 300 K if the intrinsic carrier concentration is 1.5×10^{16} atoms/m^3. The sample is then doped with 10^{23} phosphorus atoms/m^3. Determine the equilibrium hole concentration, conductivity and position of the Fermi level relative to the intrinsic level.

Solution. Given

$$\mu_n = 0.135 \text{ m}^2/\text{V–s}$$
$$\mu_p = 0.048 \text{ m}^2/\text{V–s}$$
$$n_i = 1.5 \times 10^{16} \text{ m}^{-3}$$

In the case of intrinsic semiconductors,

$$n = p = n_i$$

Therefore, the conductivity is given by

$$\sigma = en_i (\mu_n + \mu_p)$$
$$= 1.6 \times 10^{-19} \times 1.5 \times 10^{16} \times (0.135 + 0.048)$$
$$= 4.39 \times 10^{-4} (\Omega\text{-m})^{-1}$$

In the extrinsic case, since $N_d >> n_i$, and assuming all the donors to be ionized, we have

$$n \cong N_d^+ = 10^{23} \text{ atoms/m}^3$$

Therefore, the equilibrium hole concentration is

$$p = \frac{n_i^2}{n} = \frac{\left(1.5 \times 10^{16}\right)^2}{10^{23}} = 2.25 \times 10^9 \ m^{-3}$$

The conductivity is given by

$$\sigma = en\mu_n$$
$$= 1.6 \times 10^{-19} \times 10^{23} \times 0.135$$
$$= 21.6 \times 10^2 (\Omega\text{-m})^{-1}$$

From Eq. (7.42),

$$E_F - E_i = kT \ln\left(\frac{n}{n_i}\right)$$

$$= 8.62 \times 10^{-5} \times 300 \ln\left(\frac{10^{23}}{1.5 \times 10^{16}}\right)$$

$$= 0.406 \text{ eV}$$

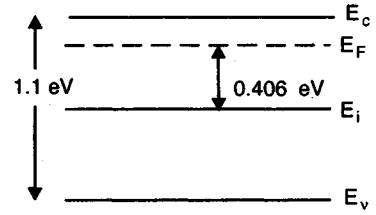


Fig. 7.11

The position of the Fermi level in the band diagram is shown in Fig. 7.11.

Example 7.2. In intrinsic GaAs, the electron and hole mobilities are 0.85 and 0.04 m^2/V-s respectively and the corresponding effective masses are 0.068 m_0 and 0.5 m_0 respectively where m_0 is the rest mass of an electron. Given the energy band gap at 300 K as 1.43 eV, determine the intrinsic carrier concentration and conductivity.

Solution. Given,

$$\mu_n = 0.85 \text{ m}^2\text{/V-s}$$
$$\mu_p = 0.04 \text{ m}^2\text{/V-s}$$
$$m_n^* = 0.068\ m_o$$
$$m_p^* = 0.5\ m_o$$
$$E_g = 1.43 \text{ eV}$$
$$T = 300 \text{ K}$$

From Eq. (7.36), the intrinsic carrier concentration is given by

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} \left(m_n^* m_p^*\right)^{3/4} \exp\left(-\frac{E_g}{2kT}\right)$$

$$= 2\left[\frac{2\pi \times 1.38 \times 10^{-23} \times 300}{(6.63 \times 10^{-34})^2}\right]^{3/2} \times$$

$$\times \left[0.068 \times 0.5 \times \left(9.1 \times 10^{-31}\right)^2\right]^{3/4} \exp\left(-\frac{1.43}{2 \times 8.62 \times 10^{-5} \times 300}\right)$$

$$= 1.94 \times 10^{12} \text{ m}^{-3}$$

$$= 1.94 \times 10^{12} \text{ m}^{-3}$$

The conductivity is given by

$$\sigma = en_i (\mu_n + \mu_p)$$
$$= 1.6\times10^{-19}\times1.94\times10^{12}\times(0.85+0.04)$$
$$= 2.76\times10^{-7} (\Omega\text{-m})^{-1}$$

Example 7.3. The resistivity of an intrinsic semiconductor is 4.5 ohm-m at 20°C and 2.0 ohm-m at 32°C. What is the energy band gap ?

Solution. For an intrinsic semiconductor, we have

$$\sigma = en_i\mu$$

Using Eq. (7.36), we get

$$\sigma = 2e\mu \left(\frac{2\pi kT}{h^2}\right)^{3/2} \left(m_n^* m_p^*\right)^{3/4} \exp\left(-\frac{E_g}{2kT}\right)$$

$$= CT^{3/2} \exp\left(-\frac{E_g}{2kT}\right)$$

where C is a constant given by

$$C = 2e\mu \left(\frac{2\pi k}{h^2}\right)^{3/2} \left(m_n^* m_p^*\right)^{3/4}$$

If σ_1 and σ_2 are the conductivities at temperatures T_1 and T_2 respectively, then

$$\frac{\sigma_1}{\sigma_2} = \frac{\rho_2}{\rho_1} = \left(\frac{T_1}{T_2}\right)^{3/2} \exp\left\{\frac{E_g}{2k}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right\}$$

where ρ represents the resistivity. The above equation can also be expressed as

$$\frac{E_g}{2k}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) = ln\frac{\sigma_2}{\sigma_1} + \frac{3}{2} ln\frac{T_1}{T_2}$$

Now, we have

$T_1 = 20 + 273 = 293$ K, $\quad \rho_1 = 1/\sigma_1 = 4.5$ ohm-m

$T_2 = 32 + 273 = 305$ K, $\quad \rho_2 = 1/\sigma_2 = 2.0$ ohm-m

$$\therefore \quad \frac{E_g}{2\times 1.38\times 10^{-23}}\left(\frac{1}{293}-\frac{1}{305}\right) = \ln\left(\frac{4.5}{2.0}\right) + \frac{3}{2}\ln\left(\frac{293}{305}\right)$$

or
$$E_g = 1.54 \times 10^{-19} \text{ J}$$
$$= 0.96 \text{ eV}$$

SUMMARY

1. Semiconductors are materials which have electrical conductivity lying between those of conductors and insulators. They exhibit band gaps of the order of 1 eV and negative temperature coefficient of resistance. Examples of these materials are Si, Ge, GaAs, InP, CdS, etc.

2. In a semiconductor, two types of current carriers are present, viz., electrons and holes. Electrons conduct in the conduction band and holes conduct in the valence band.

3. In a pure or intrinsic semiconductor, electrons and holes are thermally generated and are equal in number.

4. In a doped or extrinsic semiconductor, the conduction is mainly by either electrons (n-type conduction) or holes (p-type conduction). The electrons in the n-type and holes in the p-type semiconductors are called majority carriers. The holes in the n-type and electrons in the p-type material are termed the minority carriers.

5. Mobility is the velocity acquired by a carrier in a unit electric field. Electrons have greater mobility than holes. Hence devices with n-type conduction are mostly preferred to those with p-type conduction.

6. In general, two types of currents flow in a semiconductor — the drift current and the diffusion current. The motion of carriers in an electric field constitutes the drift current. The diffusion current arises from the motion of carriers under the effect of concentration gradient of the carriers. In a uniformly doped semiconductor, the latter contribution is absent.

7. The conductivity of a semiconductor, in general, is given by

$$\sigma = e(n\mu_n + p\mu_p)$$

8. The conductivity of an intrinsic semiconductor increases with temperature. The plot of ln σ versus $1/T$ is a straight line and can be used to determine the band gap.

9. The concentration of electrons in the conduction band and holes in the valence band are given by the expressions

$$n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} exp\left[-\left(\frac{E_c - E_F}{kT}\right)\right]$$

$$p = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} exp\left[-\left(\frac{E_F - E_v}{kT}\right)\right]$$

The intrinsic carrier concentration is

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} \left(m_n^* m_p^*\right)^{3/4} exp\left(-\frac{E_g}{2kT}\right)$$

10. The law of mass action, i.e.,

$$np = n_i^2$$

holds for both intrinsic and extrinsic semiconductors.

11. In an intrinsic semiconductor, the Fermi level lies in the middle of the conduction band and the valence band edges. At moderate temperatures and impurity concentrations, the Fermi level of an *n*-type semiconductor lies almost in the middle of the donor level and the conduction band edge, whereas in a p-type semiconductor, it lies in the middle of the acceptor level and the valence band edge. As the temperature increases, the Fermi level approaches the intrinsic level.

VERY SHORT QUESTIONS

1. What is an intrinsic semiconductor?
2. What is an extrinsic semiconductor?
3. Explain the concept of hole.
4. What type of carriers is present in a semiconductor?
5. Define mobility of a charge carrier.
6. What is doping?
7. Draw the energy level diagram for an *n*-type semiconductor and label it.
8. Draw the enrgy level diagram for a *p*-type semiconductor and label it

9. What are majority carriers and minority carriers?
10. Define law of mass-action. For what type of semiconductors does it hold?
11. Define drift velocity of a carrier?
12. What are degenerate semiconductors?
13. Which has greater mobility, electron or hole?
14. Can a semiconductor containing both *n*-type and *p*-type impurities behave as intrinsic semiconductor? Give reason.

SHORT QUESTIONS

1. Give the properties of holes vis-a-vis electrons.
2. Explain the conduction mechanism for *n*-type and *p*-type semiconductors.
3. What are donors and acceptors? Give two examples of each.
4. Derive and discuss the law of mass action.
5. Explain the concepts of drift current and diffusion current. How are they different?
6. Obtain an expression for conductivity of an intrinsic semiconductor. How does it vary with temperature?
7. Show a typical variation of conductivity with temperature for an extrinsic semiconductor and explain the different regions.
8. Define the Fermi level. What is its importance in electronic grade materials?

LONG QUESTIONS

1. Discuss the current conduction in semiconductors. How do conductivity of a semiconductor and a metal change with impurity content? Explain the difference in behaviour of these two materials due to change in conductivity.
2. Derive expression for density of free electrons and holes in an intrinsic semiconductor. Show that the Fermi level lies halfway between the valence band and the conduction band.
3. What is an extrinsic semiconductor? Discuss the variation of the Fermi level with temperature for an *n*-type semiconductor.

4. Derive an expression for density of electrons in the conduction band for an *n*-type semiconductor.
5. What are mobility and conductivity? Obtain an expression for conductivity of doped semiconductors.
6. Derive expressions for electron and hole concentrations for an intrinsic semiconductors. Use these results to obtain intrinsic carrier concentration.
7. Show that the product of electron and hole concentrations in a semiconductor is constant at a given temperature. How is the energy gap determined from the measurement of electrical coductivity of a semiconductor?

PROBLEMS

1. An intrinsic germanium crystal has a hole density of 10^{19} m^{-3} at room temperature. When doped with antimony, the hole density decreases to 10^{17} m^{-3} at the same temperature. Calculate the majority carrier density. (10^{21} m^{-3})
2. The band gaps of diamond and silicon are 5.4 and 1.1 eV respectively. Estimate the temperature at which diamond has the same conductivity as Si at 27°C. (1200°C)
3. The conductivity of intrinsic Si is 4.17×10^{-5} and 4×10^{-4} $(\Omega\text{-m})^{-1}$ at 0°C and 27°C respectively. Determine the average band gap of Si. (1.11 eV)
4. The conductivity of *n*-type germanium semiconductor is 39 $\Omega^{-1}m^{-1}$. If the mobility of electrons in germanium is 0.39 m^2 $V^{-1}s^{-1}$, then find the concentration of the donor atoms.

   (6.25×10^{20} m^{-3})
5. How many donor atoms should be added per cubic metre of pure germanium crystal such that an *n*-type semiconductor is formed whose conductivity is 500 mho/cm. Given that the mobility of electrons in *n*-type semiconductor is 0.385 $m^2V^{-1}s^{-1}$. (8.12×10^{24})
6. The conductivity of a semiconductor changes when the concentration of electrons is varied by changing the position of impurity level. Show that it passes through a minimum when the concentration of electrons becomes $n_i\sqrt{\mu_p/\mu_n}$ where n_i is the intrinsic carrier concentration, μ_n and μ_p represent the mobilities of electrons and holes respectively. Dtermine the minimum value of conductivity.($2ne\mu_n$)

7. The mobilities of electrons and holes in intrinsic Ge are 0.39 and 0.19 m^2/V-s respectively. Determine the intrinsic carrier concentration and conductivity of Ge at 300 K if the band gap of Ge is 0.67 eV and the effective masses of electrons and holes are $0.55m_0$ and $0.37m_0$ respectively, m_0 being the electronic rest mass. How many dopants must be added per cubic metre of Ge to increase its conductivity by a factor of 10^4? ($1.81\times10^{19}m^{-3}$, 1.68 Ω^{-1}-m^{-1}, $2.69\times10^{23}m^{-3}$)

8. In an *n*-type semiconductor, the Fermi level lies 0.5 eV below the conduction band. If the concentration of donor atoms is tripled, find the new position of the Fermi level, given kT = 0.03 eV.

(0.48 eV below the conduction band)